

Runbook: Production Database Point-in-Time Restore

Meridian Commercial Bank - Internal Document

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Purpose

Describes the process for restoring a production database to a specific point in time, used for both disaster recovery and data-correction scenarios (for example, recovering from a bad batch job or data corruption event).

Preconditions

- Confirm the target restore time with the requesting engineer and, for any restore affecting customer-facing data, with the Database Platform Lead.
- Confirm whether a fresh restore (new instance) or in-place restore is required. In-place restores cause downtime and require a maintenance window; fresh restores can run alongside production.

Procedure (Fresh Restore, Preferred)

1. Identify the nearest snapshot before the target restore time in the backup catalog: `db-cli backups list --db <db-name>`
2. Provision a new instance from that snapshot: `db-cli restore --snapshot <snapshot-id> --target-instance <new-instance-name>`
3. Apply write-ahead log (WAL) replay to reach the exact target timestamp: `db-cli wal-replay --instance <new-instance-name> --until <timestamp>`
4. Validate data integrity against expected row counts / checksums provided by the requesting team.
5. If this is a disaster-recovery failover (not a data-correction restore), promote the new instance to primary following the "Primary Promotion" section of this runbook and update connection strings in the service configuration.

Procedure (In-Place Restore, Requires Downtime)

1. Schedule and announce a maintenance window; this always requires sign-off from the Database Platform Lead and the affected service owner.
2. Stop write traffic to the affected database (drain connections, put dependent services into maintenance mode).
3. Follow steps 1-4 above but restore into the existing instance.
4. Resume write traffic and monitor error rates for 30 minutes.

Rollback

If validation fails after a fresh restore, simply discard the new instance; production is unaffected. If validation fails after an in-place restore, restore again from an earlier snapshot and extend the maintenance window.

Escalation

- Database Platform on-call (PagerDuty schedule: db-platform-primary)

- For restores affecting the Card Authorization Service or Wire Transfer Service specifically, also page Payments Platform on-call for coordination, referencing INC-2025-10-02-002 as a precedent where replication/restore timing directly affected payments availability.